DentaMind AI: AI-Assisted Dental X-ray Analysis System

Tooth detection, FDI numbering, segmentation, finding classification and validated draft reports for panoramic radiographs, with human review

B.Tech Generative AI MDM Capstone Project · Department of Computer Science and Engineering (AI & ML)

Pimpri Chinchwad College of Engineering, Pune · 2026-27

Group ID: _____ Students: _____ (PRN _____)

Guide: Prof. _____

Research prototype: not clinically validated, not for diagnosis

Introduction and motivation

- Panoramic radiographs (OPGs) show all teeth in one image and are used for screening and treatment planning
- Reading them is time-consuming, and subtle findings such as early caries or small periapical lesions are easy to miss
- Deep learning detects and numbers teeth well, but finding-level classification is harder and published systems rarely show their uncertainty
- Generative AI can turn structured model output into readable reports, but free text can state things the image does not support
- **Goal of DentaMind AI:** a transparent assistant that shows evidence, flags doubt, and leaves the decision to a person



X-ray Analysis page: 27 regions on DENTEX test image val_0

Problem statement

- Build an end-to-end system that, given a panoramic radiograph, locates and numbers each tooth (FDI), outlines it, and suggests the closest of four findings: Caries, Deep Caries, Impacted, Periapical Lesion
- Generate a structured, multilingual (English, Hindi, Marathi) draft report that contains only statements supported by the model evidence
- Make uncertainty visible and route doubtful regions to a human reviewer
- **Evaluate honestly:** separate validation from test data, avoid data leakage, and never promote a model on test-set results
- **Constraints:** public DENTEX data only (no healthy-tooth labels, no patient IDs), one 4 GB laptop GPU, local deployment

Objectives

- **O1:** Audit and de-duplicate the DENTEX data and build leakage-free train / validation / calibration / test splits
- **O2:** Train tooth detection, FDI numbering, instance segmentation and finding classification models
- **O3:** Combine them in one inference pipeline with flip-TTA uncertainty, review flags and abstention
- **O4:** Generate evidence-grounded draft reports in three languages with an automatic output validator
- **O5:** Provide a web interface for analysis, reports and human review, with an auditable review trail
- **O6:** Evaluate with precision, recall, F1, AP, Dice and calibration; test class-imbalance remedies under a pre-registered rule

Literature review

Work	Task	Approach	Gap relevant to us
Lee et al., 2018 [14]	Caries, periapical films	GoogLeNet Inception v3 CNN	Single task; no tooth numbering or report
Tuzoff et al., 2019 [15]	Tooth detection + numbering	Faster R-CNN + CNN classifier	No pathology
Chen et al., 2019 [16]	Detection + numbering, periapical films	Faster R-CNN + rule-based post-processing	No findings or report
Ekert et al., 2019 [17]	Apical lesions, panoramic	Deep CNN	Single finding; no human-review loop
Hamamci et al., 2023 [11], [8]	DENTEX: enumeration + diagnosis	Hierarchical diffusion-based detector	Benchmark scores; no calibrated abstention or reporting
Jing et al., 2018 [5]	Medical report generation	Co-attention + hierarchical LSTM	Free text can be unsupported by evidence
Stephan et al., 2024 [6]	ChatGPT for OPG reports	LLM-assisted reporting study	Reports omitted key diagnostic details

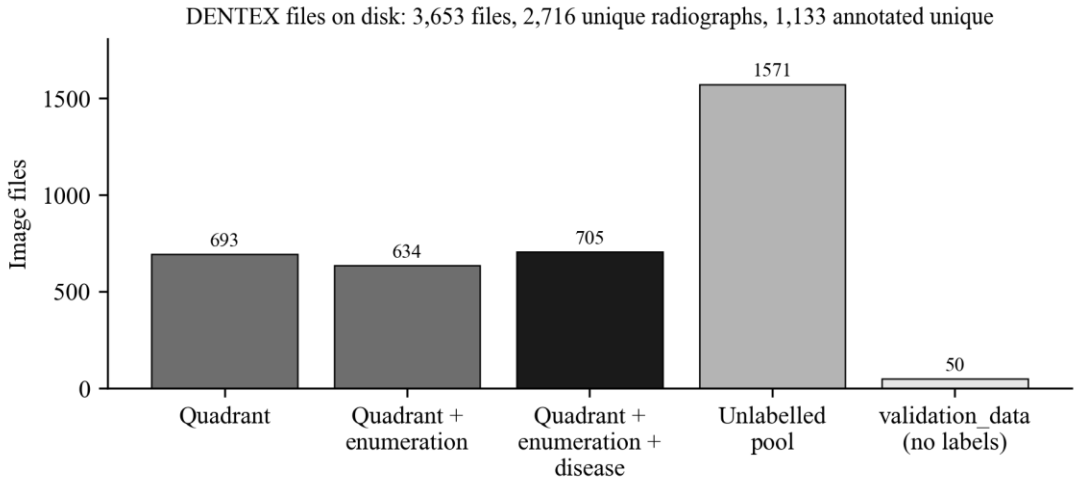
Common pattern: strong single-task accuracy, but little attention to uncertainty, leakage control, or keeping generated text faithful to the evidence.

Research gap

- **Integration:** tooth detection, numbering, segmentation and findings are usually studied separately
- **Uncertainty:** closed-set classifiers always output a class, so a system needs abstention and review flags
- **Faithful generation:** LLM reports can omit or invent findings, so output must be checked against the model evidence
- **Evaluation hygiene:** duplicates across DENTEX folders, test-set reuse and selection bias are rarely audited
- **Imbalance:** rare classes (Periapical Lesion is about 7 % of findings) are often reported only through accuracy
- DentaMind addresses these with one pipeline, a validator, a review workflow, a leakage audit and pre-registered experiments

Dataset: DENTEX (MICCAI 2023 challenge)

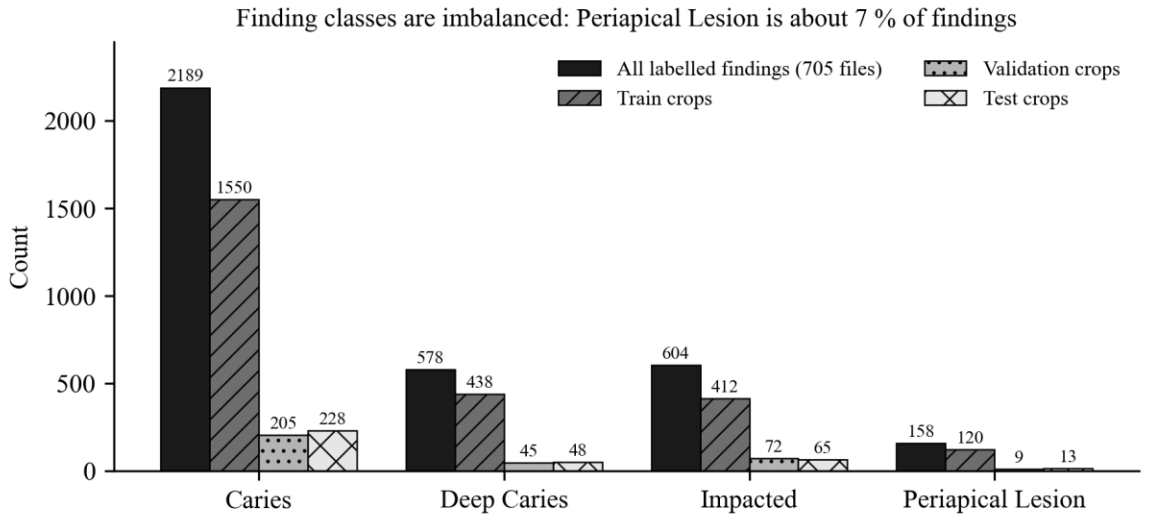
- **3,653 PNG files in five folders:** 2,716 unique radiographs after pixel-hash de-duplication
- 1,133 unique annotated radiographs; 1,598 unlabelled files; 50 unlabelled 'validation_data' files used as demo images
- **Three label levels:** quadrant (693), quadrant + enumeration (634, 18,095 tooth polygons), + disease (705, 3,529 findings)
- **Finding labels:** Caries 2,189 · Deep Caries 578 · Impacted 604 · Periapical Lesion 158
- No healthy-tooth label and no patient identifier



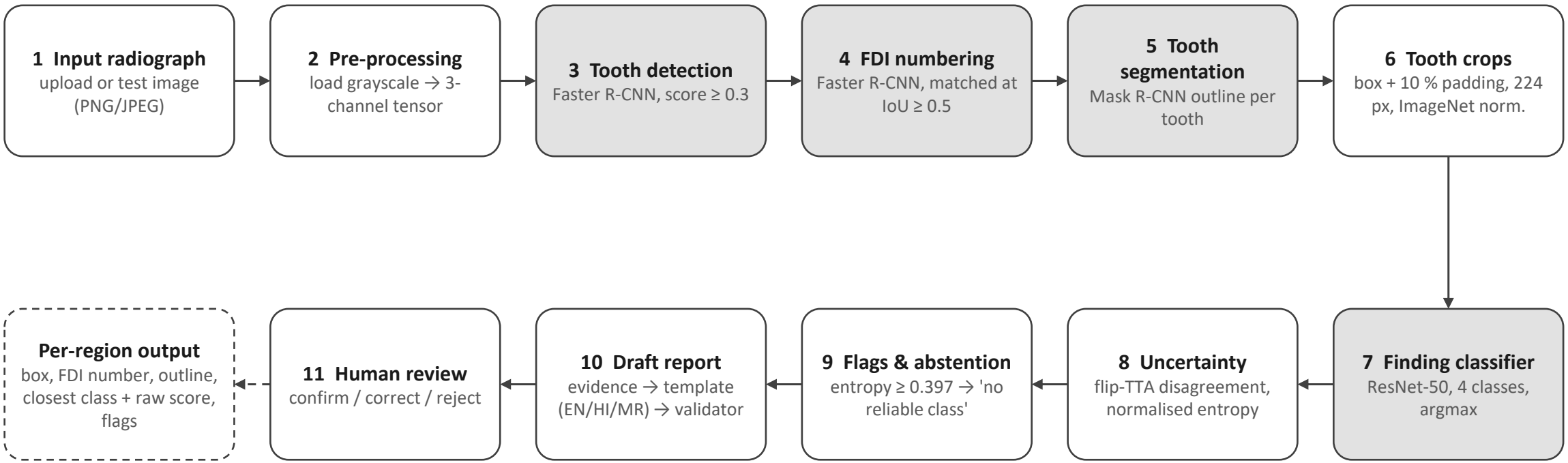
638 groups of duplicate files exist across folders, so splitting by file name would leak films between train and test.

Splits and class imbalance

- **Duplicate-aware split per unique radiograph:** train 794 · validation 113 · calibration 113 · test 113
- **Classifier crops (train):** Caries 1,550 · Deep Caries 438 · Impacted 412 · Periapical Lesion 120
- **Validation crops:** 331 (205 / 45 / 72 / 9); test crops: 354 (228 / 48 / 65 / 13)
- Only 9 validation and 13 test Periapical Lesion examples, so rare-class metrics are very noisy
- **Leakage audit:** 18 automated checks, 17 pass, 1 informational warning

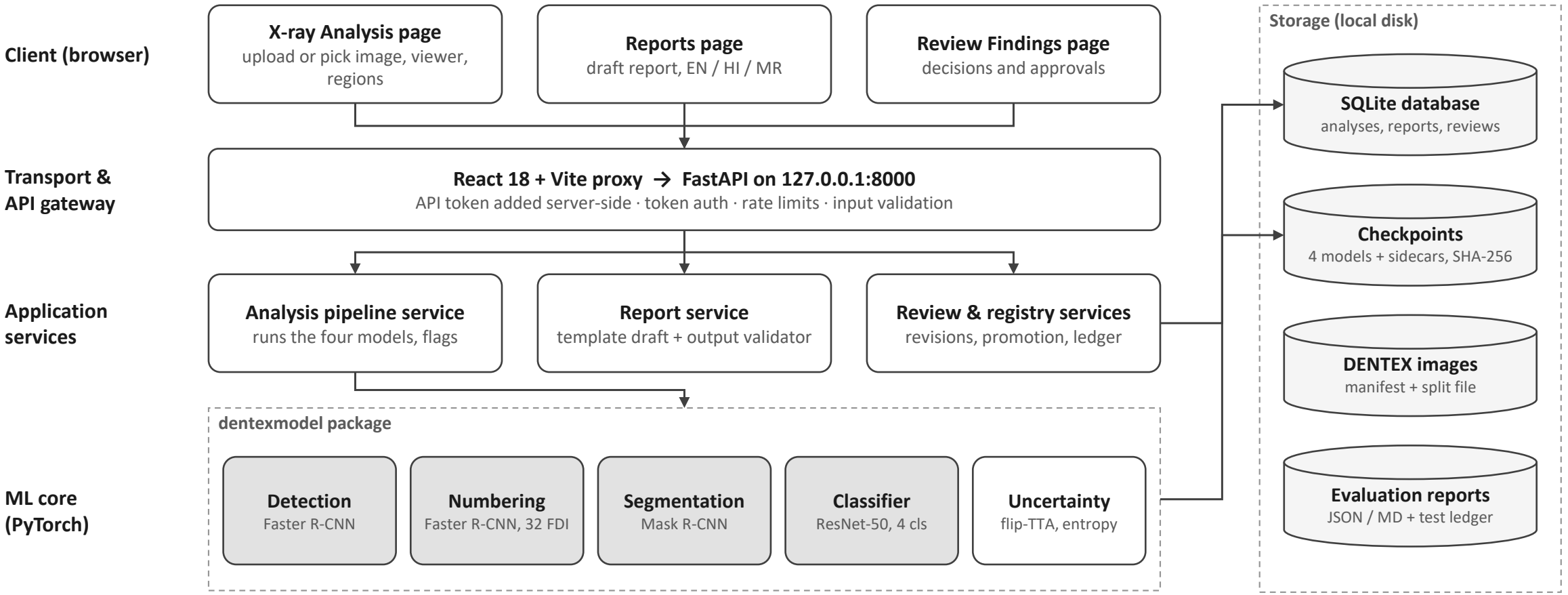


Methodology: AI inference pipeline



Order of operations as implemented in backend/services/pipeline.py. Thresholds: detection score 0.3 (boxes below 0.5 are flagged as weak detections); abstention when normalised entropy ≥ 0.3966, chosen by Youden's J on the calibration split; flip-instability flag when TTA disagreement ≥ 0.0762. The classifier has no healthy class, so every detected tooth receives its closest of four findings; the system therefore presents a candidate for review, never a diagnosis.

System architecture



Grey boxes are trained networks; white boxes are ordinary code. Everything runs on one machine and the API listens only on the loopback address. An optional OpenAI-compatible LLM path for report wording exists in code but is not configured, so all drafts in this project come from the deterministic template.

Technology stack

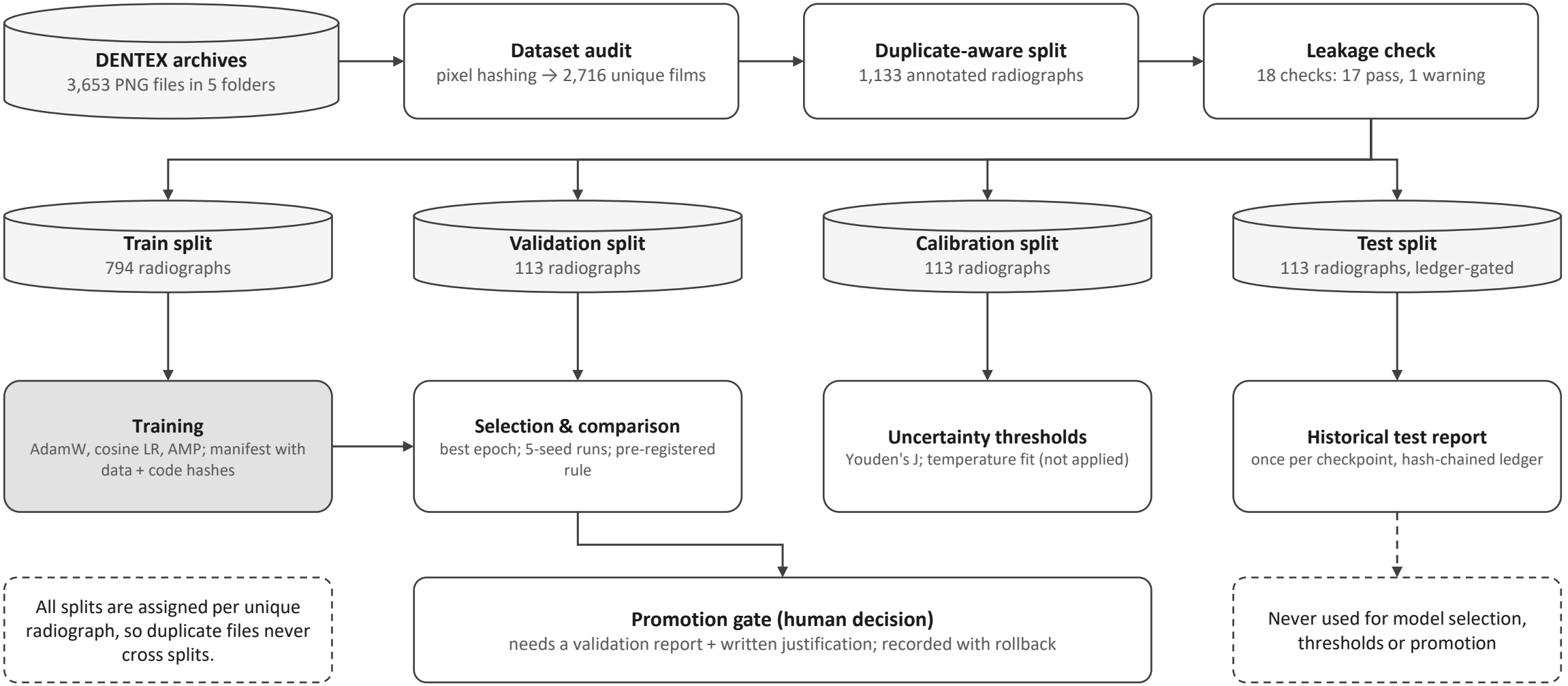
Layer	Technology	Role in DentaMind
Models	PyTorch 2.6, torchvision 0.21 (CUDA 12.4)	Faster R-CNN, Mask R-CNN, ResNet-50; training and inference
Backend	Python 3.13, FastAPI 0.143, Uvicorn	REST API, token auth, rate limits, pipeline and report services
Data	NumPy, Pillow, SQLite	Image loading, hashing, crops; analyses, reports and reviews
Frontend	React 18, Vite 5, Tailwind CSS, Recharts	Analysis viewer, reports, review queue, research pages
Reports	Template engine (EN / HI / MR) + validator	Evidence-grounded drafts; optional LLM path not configured
Quality	pytest (233 tests), scripts for leakage and ledger	Regression tests, integrity checks, reproducible evaluation
Hardware	NVIDIA RTX 3050 Laptop GPU, 4 GB	All training, evaluation and inference

Implementation: trained models

Model	Architecture	Training data	Schedule
Tooth detection	Faster R-CNN R50-FPN, COCO-pretrained, 1 class	440 train radiographs (enumeration set)	12 epochs, batch 2, AdamW 1e-4, best epoch 9
FDI numbering	Faster R-CNN R50-FPN, 32 FDI classes	440 train radiographs	16 epochs, batch 2, AdamW 1e-4
Tooth segmentation	Mask R-CNN R50-FPN, polygon masks	440 train radiographs	8 epochs, batch 1, AdamW 1e-4
Finding classifier	ResNet-50, ImageNet-pretrained, 224 px crops	2,520 crops from 476 radiographs	15 epochs, batch 32, AdamW 3e-4, class weights

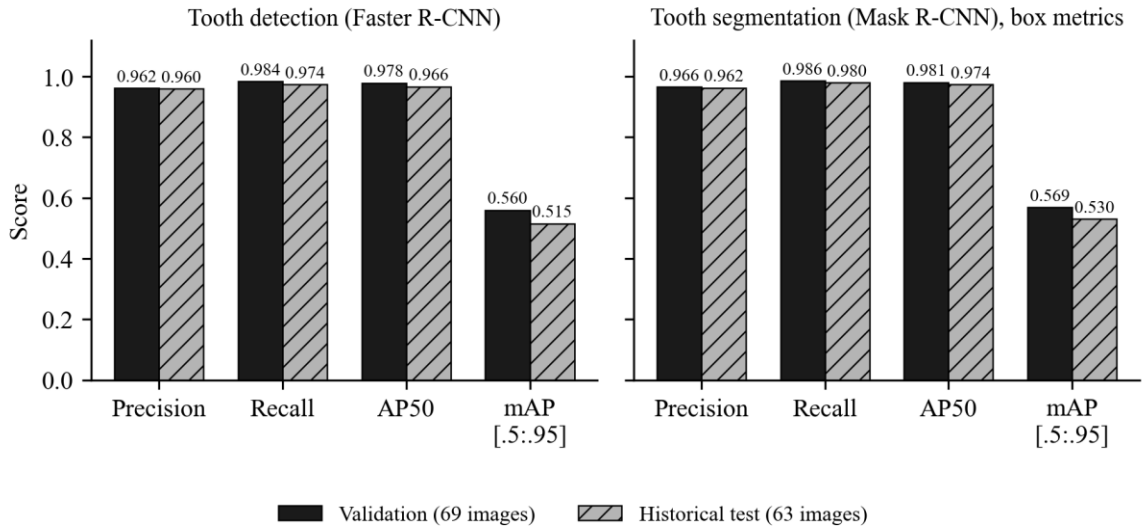
- Every checkpoint has a JSON sidecar (classes, data counts, history) and a SHA-256 hash; new checkpoints also get a training manifest
- Selection always used the validation split (best epoch by mAP or macro-F1); the test split was never used for selection
- Reports are drafted from structured evidence and checked by a validator before they are shown

Training and evaluation workflow



Results: detection and segmentation

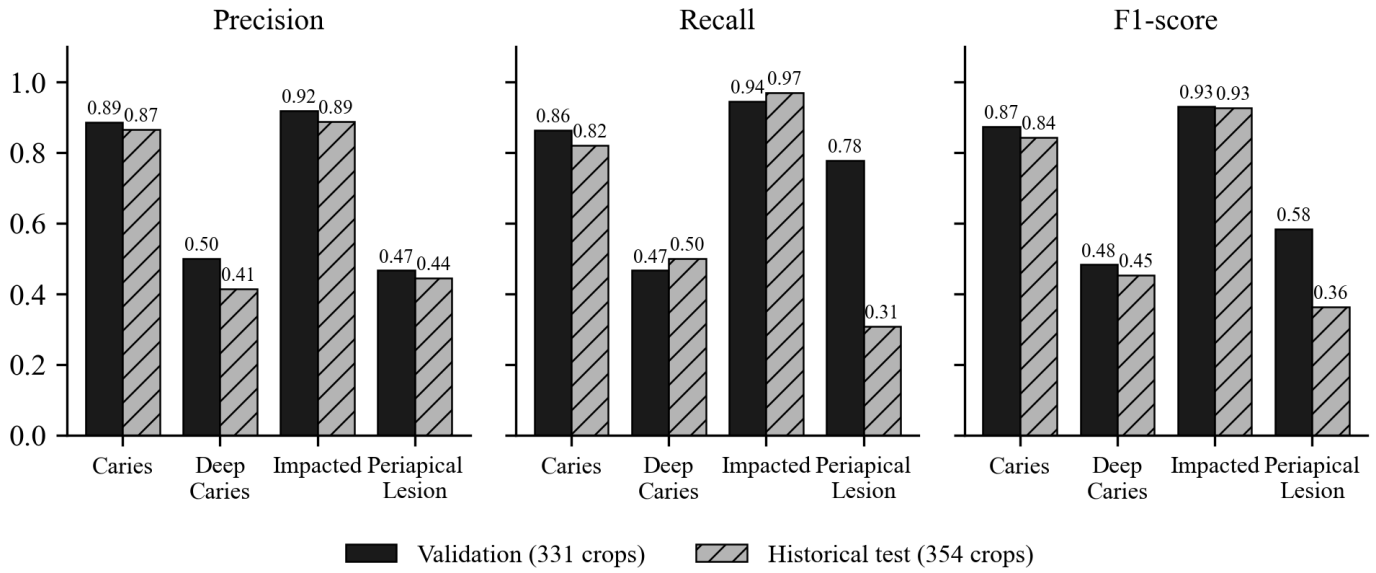
- **Detection, validation:** precision 0.962, recall 0.984, AP50 0.978, mAP 0.560
- **Detection, historical test:** precision 0.960, recall 0.974, AP50 0.966, mAP 0.515
- **Segmentation, historical test:** mask AP50 0.974, mean Dice 0.873, mean IoU 0.780 (matched teeth)
- **Numbering:** exact FDI accuracy 93.6 % (validation) and 96.8 % (test) among matched teeth; quadrant accuracy 99.7 % / 99.9 %
  - **Operating point:** score 0.3, IoU 0.5. mAP averages IoU 0.5 to 0.95, so it is lower: box placement is good but not pixel-tight



Results: finding classification (ground-truth boxes)

Split	Accuracy	Macro precision	Macro recall	Macro F1	Crops
Validation	0.825	0.693	0.763	0.718	331
Historical test	0.785	0.653	0.649	0.646	354

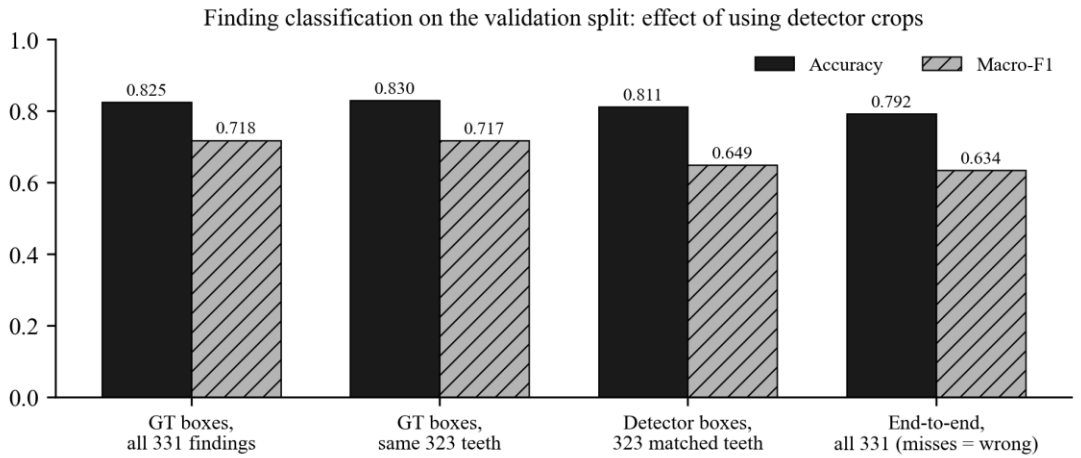
- **Strong:** Impacted (F1 0.93 / 0.93) and Caries (0.87 / 0.84)
- **Weak:** Deep Caries (0.48 / 0.45), often confused with Caries
- **Periapical Lesion:** F1 0.58 on 9 validation crops, 0.36 on 13 test crops
- Test macro ROC-AUC 0.910



Accuracy hides the rare classes: macro-F1 weights all four findings equally and is our primary metric.

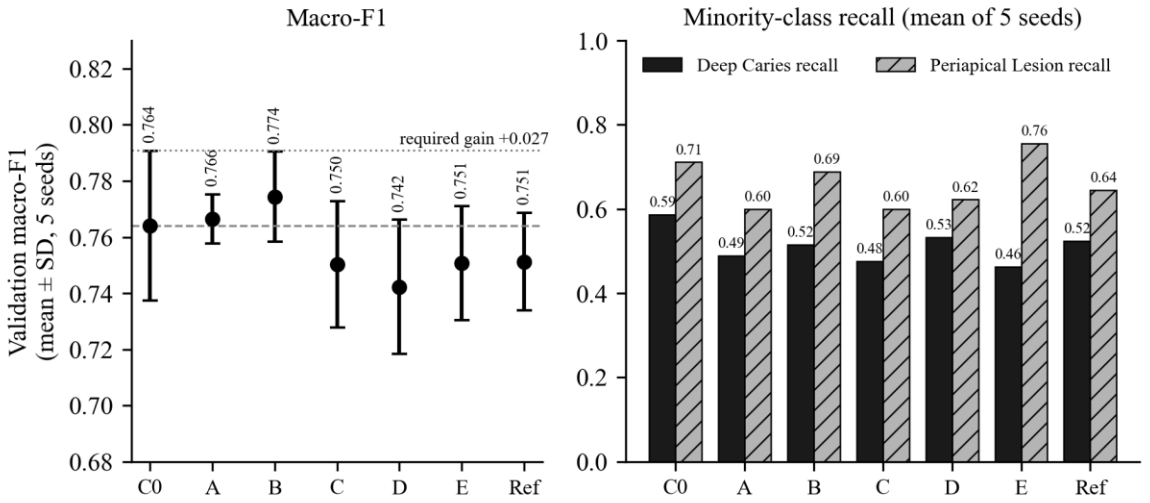
Results: ground-truth boxes vs detector boxes (validation)

- **Same 323 teeth:** ground-truth crops give accuracy 0.830 and macro-F1 0.717; detector crops give 0.811 and 0.649
- End-to-end over all 331 findings (misses count as wrong): accuracy 0.792, macro-F1 0.634
- **Detector on 24 fully annotated images:** 716 of 719 teeth found, 17 false positives (P 0.977, R 0.996)
- **Crop sensitivity:** about 10 % of predictions change even when detector and ground-truth boxes overlap at IoU above 0.9
- 1,580 of 1,903 detected teeth have no finding label but still receive a class (no healthy class)



Class-imbalance experiments

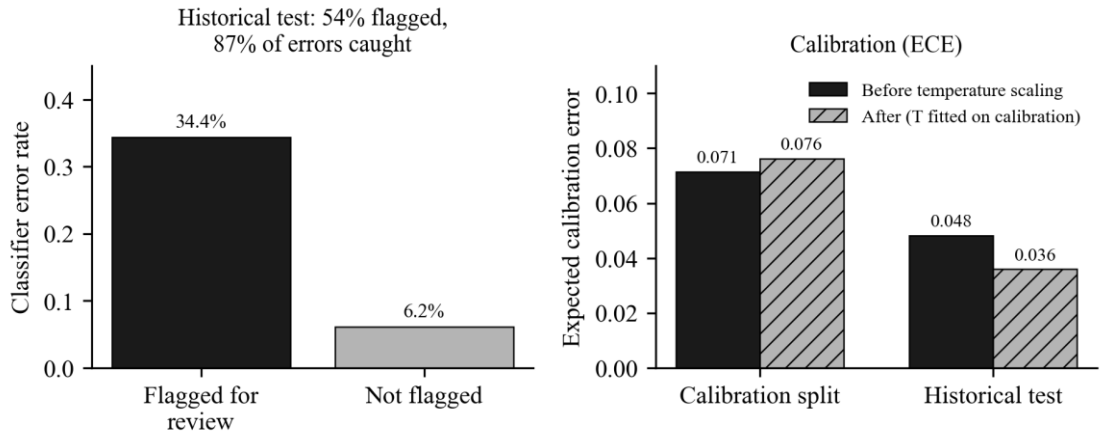
- **Pre-registered rule:** paired macro-F1 gain above 0.027 in at least 4 of 5 seeds, without lowering minority recall
- A weighted sampler +0.002 (2/5) · B oversampling +0.010 (3/5) · C focal loss -0.014 (0/5)
- D sqrt weights -0.022 (1/5) · E minority-only augmentation -0.013 (1/5)
- **No variant passed:** A to D gained precision but lost Deep Caries and Periapical Lesion recall; E lost Deep Caries recall in 5/5 seeds
- **Threshold tuning:** +0.023 in-sample, but -0.005 under cross-fitting, so argmax was kept



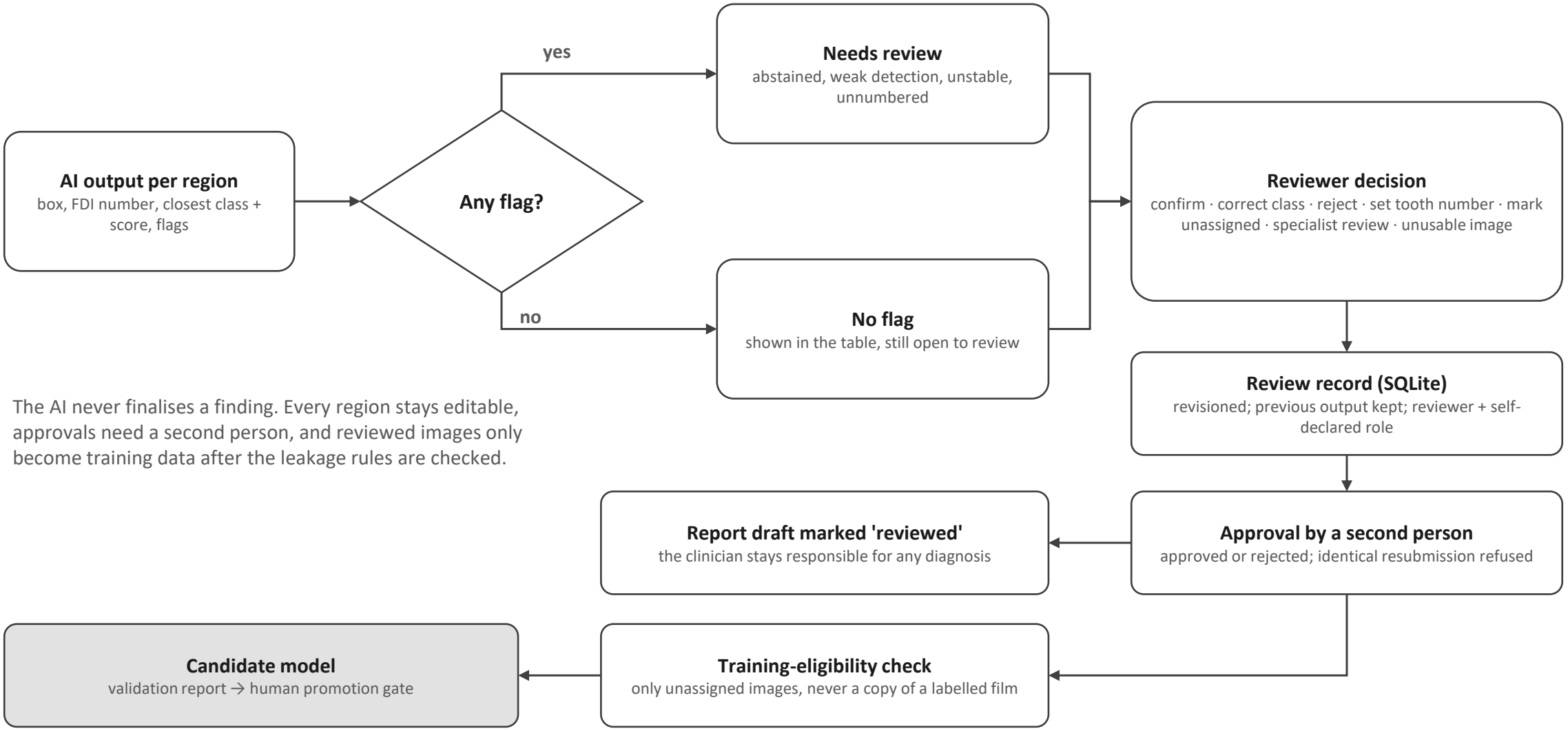
Not promoted: no gain beyond seed-to-seed noise (SD 0.027), and the minority classes became worse.

Uncertainty, abstention and recipe study

- Flip-TTA plus normalised entropy; review threshold 0.397 chosen by Youden's J on the calibration split
- **Historical test:** 54 % of crops flagged; they contain 87 % of the errors (error rate 34 % flagged vs 6 % not flagged)
- ECE 0.048 on test; temperature T = 0.95 (fitted on calibration) is reported but not applied
- **Recipe study:** the production trainer reproduces the deployed classifier bit-for-bit (seed 42); its five seeds give detector-crop macro-F1 0.722 ± 0.045, and the deployed run is the weakest (0.651)
- The alternative recipe (0.724 ± 0.015) did not pass the rule either (10 of 25 cross pairs), so nothing was promoted



Human-review workflow



The AI never finalises a finding. Every region stays editable, approvals need a second person, and reviewed images only become training data after the leakage rules are checked.

Limitations

- **No healthy class:** every detected tooth receives one of four findings; on test image val_0, 18 of 27 regions abstained
- **Rare classes:** Deep Caries recall 0.33 and Periapical Lesion 4/9 end-to-end on validation; tiny support makes these estimates unstable
- **Crop sensitivity:** small box shifts change about 10 % of predictions; the detector-box classifier is weaker than the ground-truth-box classifier
- **No independent external validation:** one public dataset, no patient IDs (patient-level separation unproven), no other scanners or sites
- The test split was used historically; thresholds and the LLM path are provisional; Hindi and Marathi text was not reviewed by dental professionals
- Not clinically validated, not a medical device, not for diagnosis

Conclusion and future work

- **Built a working local system:** detection (validation recall 0.984), numbering (93.6 % exact), segmentation (test Dice 0.873) and a four-class finding classifier
- Classification is the weak link (validation macro-F1 0.718; 0.634 end-to-end), and we report this openly instead of tuning on the test split
- Reports are generated only from evidence and checked by a validator; humans stay in control through review and approval
- **Future:** add healthy and 'other' labels (open-set), collect more Periapical Lesion and Deep Caries cases, and train on detector-style crops
- **Future:** external multi-centre validation with patient-level splits, a reader study with dentists, and a validated LLM wording layer
- **Future:** calibrated probabilities, tooth-level heatmaps, and DICOM / PACS integration

Key references and thank you

- [11] I. E. Hamamci et al., DENTEX benchmark, arXiv:2305.19112, 2023
- [8] I. E. Hamamci et al., Diffusion-based hierarchical multi-label object detection, MICCAI 2023, LNCS 14225
- [18] S. Ren et al., Faster R-CNN, NeurIPS 2015 · [19] K. He et al., Mask R-CNN, ICCV 2017 · [20] K. He et al., ResNet, CVPR 2016
- [26] C. Guo et al., On calibration of modern neural networks, ICML 2017
- [27] G. Wang et al., Test-time augmentation uncertainty, Neurocomputing 335, 2019
- [12] S. Kaufman et al., Leakage in data mining, ACM TKDD 6(4), 2012

- **Full IEEE reference list:** Project Report, References chapter
- Thank you. Questions?